THEMIS

Trusted Human-AI Enablement for Matter Intelligence and Safety

Addendum A | Five New Services + Three Architectural Extensions

DOCUMENT TYPE

Platform Architecture Addendum

STATUS

Draft — Internal Review

EXTENDS

THEMIS Platform Architecture and Service Design | 9 services | April 25, 2026 — addendum adds 5 services and 3 architectural extensions

Addendum Overview

This addendum extends the THEMIS Platform Architecture in two ways: five new services addressing gaps in citation verification, model drift, adversarial input, media authenticity, and document provenance; and three architectural extensions to existing services addressing gaps identified through red team analysis — cryptographic attestation of the provenance chain, corpus quality weighting in the ground truth feedback loop, and ground truth corpus temporal validity.

The five new services are CVS/VERITAS (citation verification), MDS/KRONOS (model drift), IAS/SCUDO (adversarial input screening), MAS/EIDOLON (media authenticity), and DPS/CODEX (document provenance). The three architectural extensions are: cryptographic event signing applied to Provenance/MOIRAI and all event-writing services; correction confidence weighting and human review queue in FGTS/ALETHEIA with a corresponding calibration weight function in TCS/MIMIR; and TVS/KAIROS scope extension to cover the FGTS ground truth corpus alongside active matter evidence. None of the three extensions require new services.

A sixth concern — temporal validity of the FGTS ground truth corpus — is addressed by extending TVS/KAIROS scope rather than standing up a new service. This extension is documented in Section 06. The cryptographic attestation extension (Section 07) and FGTS quality extension (Section 08) were added following red team analysis of the original addendum.

Code	Full name	Phase	Primary concern addressed
CVS / VERITAS	Citation Verification Service	3-4	Proactive generation-time citation resolution. Eliminates the fabricated-citation failure mode at the source.
MDS / KRONOS	Model Drift Service	3-4	Vendor model version change detection, active matter flagging, calibration re-validation, and matter-level version pinning.
IAS / SCUDO	Input Adversarial Screening Service	5-6	Ingestion-point scanning for prompt injection and adversarial instruction patterns before retrieved chunks reach the context window.
TVS / KAIROS (ext.)	Temporal Validity — scope extended	7-8	Ground truth corpus validity tracking incorporated into existing TVS service. KCS invalidation events now propagate into the FGTS corpus alongside active matter evidence.
MAS / EIDOLON	Media Authenticity Service	5-6	Authenticity risk scoring for video, audio, image, and scanned document evidence. Deepfake detection, manipulation analysis, and OCR confidence scoring before media artifacts enter the evidence corpus.
DPS / CODEX	Document Provenance Service	3-4	Document-layer chain of custody for high-volume document generation. Buffers paragraph-level provenance events to Provenance/MOIRAI. Integrates with the custom web authoring solution. Enforces PCES matter scope at document composition level.

Section 01 — CVS / VERITAS

		EPITHET	MEANING
CVS NEW	**Citation Verification Service** *VERITAS / Latin for truth — the standard against which legal citation is measured.*	**VERITAS**	Latin for truth — the standard against which legal citation is measured.

Purpose

CVS is the only THEMIS service that addresses AI failure at the generation layer rather than after the fact. Its single function is to resolve every legal citation produced by the model against authoritative legal databases before the response is displayed to the analyst. A citation that does not resolve to a real case, statute, or regulation is a fabrication — and in legal work, fabricated citations have produced bar discipline, sanctions, and client harm at peer firms.

The Verification Queue and Source Type Badges address citation quality reactively: the analyst sees output, expands the provenance panel, and can choose to verify. CVS addresses it proactively: the response does not render until every citation-format string has been checked. The analyst never sees an unresolved citation presented as confident fact.

Design Principles

- Runs at generation time, after PGS/NOMOS output screening and before the response is displayed to the analyst.
- Pattern-matches legal citation formats: case citations (Name v. Name, reporter, year), statute references (U.S.C., C.F.R., state equivalents), and regulation identifiers.
- Resolves each match against Westlaw and Lexis APIs within the privilege and matter scope established by PCES/AEGIS.
- Returns one of three statuses per citation: Verified (citation resolves to the expected source), Unresolved (source exists but characterisation cannot be confirmed without retrieval), or Not Found (no matching source located).
- Not Found is a hard interrupt. The response is held and the analyst is shown a blocking panel identifying the unresolvable citation before the response body renders. This is not a badge. It cannot be dismissed without explicit analyst acknowledgment.
- Verified and Unresolved citations surface as inline indicators adjacent to the citation text, distinct from claim confidence badges.

Failure Mode Addressed

RIS	Attorneys at peer firms have filed briefs citing cases that do not exist, generated by AI operating without citation verification. Several have faced bar discipline. Without CVS, ATHENA analysts relying on Training Only or Synthesis-sourced claims have no protection against this failure mode at

K the point of generation. The Verification Queue can catch it reactively if the analyst expands the provenance panel and checks — but that relies on analyst diligence, which is exactly what the ATHENA intervention design exists to compensate for.

Service Specification

Namespace	themis-quality (alongside FGTS/ALETHEIA and TVS/KAIROS)
Trigger	Fires on every model response before rendering. Activated by PGS/NOMOS interaction class — can be configured as mandatory for evidence_analysis and legal_drafting classes, advisory for research classes.
Input	Raw model response text from inference gateway. Matter scope and privilege classification from PCES/AEGIS.
Output	Citation verification manifest: [{citation_text, format_type, status: verified|unresolved|not_found, source_url, resolved_at}]. Published to Provenance/MOIRAI as CitationVerificationEvent.
Latency target	P95 < 2 seconds for responses containing up to 10 citations. Not Found cases are synchronous and blocking. Verified/Unresolved can be surfaced progressively as citations resolve.
External dependencies	Westlaw API, Lexis API. Credentials managed through Vault (themis-infra). API calls are scoped — no client matter content is transmitted, only citation strings.
Storage	Citation resolution cache (Redis) keyed on citation string plus jurisdiction. TTL: 24 hours for Verified, 4 hours for Unresolved, 1 hour for Not Found (to allow for database update propagation).
Fallback	If both external APIs are unavailable, CVS surfaces an advisory rather than blocking. Analyst is notified that citation verification is unavailable. Interaction is logged to Provenance/MOIRAI with verification_status: service_unavailable.

Integration with Existing THEMIS Services

Service	Relationship	Integration detail
PCES / AEGIS	Constraint	Matter scope constrains which citation databases CVS may query. Client matter citations are not transmitted — only citation strings — but the resolution scope is matter-aware.
PGS / NOMOS	Consumer / Policy	Interaction class determines CVS mode: mandatory-blocking for high-risk classes, advisory for lower-risk classes. Policy authoring UI updated to include CVS mode configuration per interaction class.
Provenance / MOIRAI	Producer	CVS publishes CitationVerificationEvent for every citation checked, including status, resolution detail, and timestamp. Citation verification state is part of the turn provenance record.
ERAS / LOGOS	Feeds into	Not Found and Unresolved citations are routed to ERAS

Service	Relationship	Integration detail
		as unsupported claim signals. ERAS incorporates CVS status into claim-level confidence assessment and reasoning completeness scoring.
FGTS / ALETHEIA	Feeds into	Not Found citations that the analyst confirms as fabrications (via Verification Queue action) are logged to FGTS as high-confidence correction events — fabricated citations are the clearest possible ground truth signal.
TCS / MIMIR	Feeds into	CVS Not Found rate per interaction class and per session is a calibration signal. High fabrication rates in a specific domain are a TCS flag for that domain's confidence scoring.

Epithet Rationale

VERITAS — Latin for truth. The standard against which every legal citation is ultimately measured. In the context of professional responsibility, a citation that is not true does not exist. The name reflects both the service function (truth verification) and the stakes (professional liability when the truth is fabricated).

Section 02 — MDS / KRONOS

MDS NEW	**Model Drift Service** *KRONOS / Greek god of time — who governs change across time.*	EPITHET **KRONOS**	MEANING Greek god of time — who governs change across time.

Purpose

Legal matters run for months. A matter opened in January and resolved in December will span multiple AI vendor model updates. TCS/MIMIR calibration baselines established at matter open may be invalid by matter close. Provenance/MOIRAI logs model version per session, but no service in the original architecture detects when a model version change has occurred, notifies affected matters, or supports matter-level version pinning for situations where session consistency is legally material.

MDS closes this gap. It monitors vendor API model version identifiers continuously, fires an event when a version change is detected, identifies all active matters where the prior version was used, and notifies TCS/MIMIR that calibration baselines for those matters require re-evaluation. Where pinning is configured for a matter, MDS enforces version stability across sessions until the matter is closed or the pin is explicitly released.

Design Principles

- Polls vendor API version endpoints on a configurable interval. Version change detection is based on model identifier comparison, not on output behavioral sampling.
- On detection of a version change, immediately fires a ModelVersionChangeEvent to Provenance/MOIRAI before any other action. The event is immutable and append-only.
- Identifies all active matter sessions using the prior model version from Provenance/MOIRAI session records.
- Sends re-validation requests to TCS/MIMIR for affected matters. Re-validation does not reset calibration baselines — it flags them as potentially stale pending review.
- Supports matter-level model version pinning. A pinned matter will not route inference calls to an updated model version without explicit release. Pinning is configured at matter open and can be enforced for matters where cross-session consistency is legally material.
- Version pin release requires AI Governance Committee acknowledgment for active matters. Automatic release is available for closed matters only.

The Version Consistency Problem in Legal Context

CONTEX

A partner prepares deposition strategy in February using an AI model version that characterises a regulatory position as settled. The same firm AI, updated to a new model version in April, characterises the same regulatory position as contested. The deposing attorney, relying on the February analysis without awareness of the model change, is not adequately prepared for the witness response. MDS does not prevent model updates. It ensures the firm knows when they

T occur and which active matters are affected.

Service Specification

Namespace	themis-intelligence (alongside KCS/ARGUS — both are external monitoring services)
Polling interval	Configurable per vendor. Default: every 4 hours. Version change detection latency target: < 8 hours from vendor deployment.
Input	Vendor API version endpoints. Active session records from Provenance/MOIRAI. Matter-level pinning configuration from AI Governance Committee.
Output	ModelVersionChangeEvent to Provenance/MOIRAI. Re-validation requests to TCS/MIMIR. Notification to AI Governance Committee dashboard. Version pin enforcement signal to inference gateway.
Pinning enforcement	Version-pinned matter requests are intercepted at the inference gateway. If the requested model version is no longer available from the vendor, MDS triggers a human escalation rather than silently routing to the current version.
Storage	Version history registry (PostgreSQL): {vendor, model_id, version_string, detected_at, active_matters_affected[]}. Immutable append-only record.
Governance surface	AI Governance Committee dashboard shows: current model versions per vendor, pending re-validation matters, active version pins, and historical version change log.

Integration with Existing THEMIS Services

Service	Relationship	Integration detail
Provenance / MOIRAI	Depends on / Feeds into	MDS reads active session records to identify affected matters. Writes ModelVersionChangeEvent as an immutable audit record. The version change is part of the provenance chain for any subsequent session on the affected matter.
TCS / MIMIR	Feeds into	MDS sends re-validation requests to TCS for affected matters. TCS flags calibration baselines as potentially stale and surfaces this in the analyst-facing calibration dashboard. Re-validation is triggered — not automatic reset.
PGS / NOMOS	Relationship	PGS policy can be configured to require explicit analyst acknowledgment of a model version change before proceeding with a session on an affected matter. This is a policy decision, not a default MDS behavior.
Inference Gateway	Feeds into	MDS enforces version pinning at the inference gateway. Pinned matters are intercepted before routing; non-pinned matters route to current version without interruption.
KCS / ARGUS	Architectural peer	Both MDS and KCS sit in themis-intelligence and follow the same pattern: external monitoring service that writes events to Provenance/MOIRAI and triggers downstream

Service	Relationship	Integration detail
		THEMIS service actions. They share the event bus infrastructure but have no direct service-to-service dependency.

Epithet Rationale

KRONOS — the Greek personification of time, distinct from Cronus the Titan. Kronos governs the passage of time and the change that comes with it. In THEMIS, MDS governs the change that AI model updates represent: the same question asked at a different point in time may produce a materially different answer. KRONOS is the service that makes that change visible, traceable, and manageable.

Section 03 — IAS / SCUDO

		EPITHET	MEANING
IAS NEW	**Input Adversarial Screening Service** *SCUDO / Italian for shield — the defensive layer at the ingestion boundary.*	**SCUDO**	Italian for shield — the defensive layer at the ingestion boundary.

Purpose

PCES/AEGIS enforces privilege at the data access layer. PGS/NOMOS governs interactions at the policy layer and screens outputs. Neither service addresses a specific failure mode that is structurally possible in legal document review: adversarial content embedded in the documents being reviewed.

When an analyst reviews an opposing party's filing, an exhibit, or a deposition transcript through ATHENA, that document's text enters the model's context window via the retrieval pipeline. If that text contains instruction-format strings, meta-prompt patterns, or known injection signatures — whether placed deliberately or appearing incidentally — those strings may manipulate the model's behavior in ways that PCES and PGS are not designed to detect. The attack surface is the space between document ingestion and context window population, which is exactly where IAS operates.

Design Principles

- Sits at the retrieval ingestion point, between document chunking and RQS/HERMES retrieval quality scoring. Every chunk passes through IAS before entering the retrieval candidate pool.
- Scans for instruction-format text patterns: imperative verb constructions directed at AI systems, role-reassignment language, system prompt override attempts, and jailbreak signature strings from the known injection taxonomy.
- Produces a risk score per chunk on a continuous scale. Score thresholds are configurable by PGS/NOMOS policy: high-risk chunks can be quarantined, flagged for analyst review, or allowed with a visible warning depending on policy configuration.
- Quarantined chunks are excluded from the retrieval candidate pool and cannot reach the context window until released by an authorized analyst or the AI Governance Committee.
- All injection events — detected, flagged, quarantined — are logged to Provenance/MOIRAI as InjectionScreeningEvents. The log is immutable. Detection does not modify the source document.
- IAS is not a semantic filter. It does not assess whether document content is correct, relevant, or privileged. Those concerns belong to RQS/HERMES and PCES/AEGIS respectively. IAS has exactly one concern: adversarial instruction patterns.

The Adversarial Input Surface in Legal Practice

RIS

A hostile party submits an exhibit in discovery that contains, embedded in a footnote or metadata field, text instructing an AI system to adopt a different role or ignore prior instructions. Attorneys reviewing that exhibit through an AI-assisted interface route the exhibit text through the model.

K Without IAS, the injected instruction enters the context window alongside the legitimate exhibit content. PCES will not catch it — it is not a privilege violation. PGS output screening may catch some effects, but not the instruction itself. IAS is the only service positioned to intercept the injection before it reaches the model.

Service Specification

Namespace	themis-quality (alongside CVS/VERITAS and FGTS/ALETHEIA — all three are quality assurance services)
Position in pipeline	Post-chunking, pre-RQS/HERMES. Runs on every chunk entering the retrieval candidate pool regardless of document source.
Detection methods	Signature matching against known injection taxonomy (maintained as a versioned ruleset). Embedding similarity against a curated injection pattern corpus. Heuristic scoring on imperative-directive linguistic structures. Detection methods are layered — all three run per chunk.
Risk scoring	Continuous 0.0 to 1.0 scale. Thresholds configurable by PGS policy: Advisory (0.3-0.6), Flagged (0.6-0.85), Quarantined (>0.85). Default thresholds provided; practice-group-specific overrides available.
Output per chunk	{chunk_id, risk_score, matched_patterns[], disposition: pass|advisory|flagged|quarantined, screened_at}
Storage	Injection event log (PostgreSQL, append-only): all screening events. Quarantine registry (Redis): active quarantine state per chunk. Injection taxonomy (versioned ruleset, Git-managed).
Taxonomy maintenance	The injection signature taxonomy is maintained by the platform engineering team with input from the Red Team Evaluator. Updates follow the PGS policy change control process and require AI Governance Committee approval for new high-threshold signatures.
False positive handling	Legitimate legal documents may contain imperative language that resembles injection patterns. Flagged chunks are visible to the analyst with their injection risk score and matched pattern description. The analyst can release a flagged chunk with an override reason. Override events are logged and reported to the AI Governance Committee.

Integration with Existing THEMIS Services

Service	Relationship	Integration detail
RQS / HERMES	Upstream gating	IAS runs before RQS/HERMES receives chunks. RQS only scores retrieval quality for chunks that IAS has passed, flagged (and not quarantined), or cleared via analyst override. Quarantined chunks are invisible to RQS.
PGS / NOMOS	Policy consumer	IAS threshold configuration is a PGS policy concern. Risk thresholds, disposition rules, and override permissions are authored in PGS and consumed by IAS at runtime. Policy changes to IAS behavior follow the standard PGS change control process.

Service	Relationship	Integration detail
Provenance / MOIRAI	Feeds into	Every IAS screening event is written to Provenance/MOIRAI as an InjectionScreeningEvent. Quarantine events, override events, and release events are all logged. The source document provenance record includes IAS disposition for every chunk derived from it.
PCES / AEGIS	Architectural complement	PCES governs privilege at the data access layer — which matter can access which corpus partition. IAS governs adversarial content at the ingestion layer — which content within an accessible document is safe to route to the model. The two services are complementary and non-overlapping in their concern.
FGTS / ALETHEIA	Feeds into	Confirmed injection events — where analyst review confirms adversarial intent — are logged to FGTS as a distinct correction signal category. This builds the injection pattern corpus over time and informs taxonomy maintenance.

Epithet Rationale

SCUDO — Italian for shield. The service is a defensive layer at the ingestion boundary, and the shield metaphor reflects its function precisely: it does not engage with the adversarial content or attempt to neutralise it. It identifies it, contains it, and prevents it from reaching the model. The name is deliberately non-mythological — in a platform of Greek epithets, a Latin/Italian term signals that this service operates at a different layer, closer to infrastructure than to analytical intelligence.

Section 04 — MAS / EIDOLON

MAS NEW	**Media Authenticity Service** *EIDOLON / Greek for phantom or apparition — the image that may deceive.*	EPITHET **EIDOLON**	MEANING Greek for phantom or apparition — the image that may deceive.

Purpose

Legal work increasingly involves AI-assisted analysis of video depositions, audio recordings, image evidence, and scanned documents. No service in the original THEMIS architecture assesses the authenticity of these media artifacts. The current platform can govern how AI reasons about media content — through ERAS/LOGOS — and can track its provenance — through Provenance/MOIRAI — but it cannot surface whether the underlying media is genuine.

MAS sits at media ingestion and produces an authenticity risk signal per artifact before it enters the THEMIS evidence corpus. It covers four distinct media types, each with its own failure mode taxonomy: video (deepfake and temporal manipulation detection), audio (voice synthesis and speaker inconsistency detection), images (manipulation, GAN artifact, and metadata analysis), and scanned documents (OCR confidence scoring and low-confidence region flagging). MAS does not adjudicate authenticity — that remains a human judgment — but it ensures authenticity signals are surfaced systematically rather than relying on analyst vigilance alone.

Failure Modes by Modality

Modality	Primary failure modes	MAS detection approach
Video	Deepfake generation. Temporal manipulation (frame removal, splice). Metadata timestamp and GPS spoofing. AI transcription errors propagated as ground truth.	Ensemble classifier: facial inconsistency scoring, temporal artifact detection, compression pattern analysis. Metadata authenticity check. Transcription flagged as derivative artifact with its own accuracy metadata.
Audio	Voice cloning and synthesis. Speaker misidentification. Spliced recordings. Background noise misinterpretation by AI.	Voice synthesis classifier. Speaker verification against reference embedding where available. Acoustic continuity analysis. Spectral anomaly detection.
Images	AI-generated or manipulated images. EXIF metadata spoofing (timestamp, GPS, device). Steganographic injection. Document image forgery.	GAN artifact detection. JPEG compression inconsistency analysis. EXIF metadata authenticity scoring. Steganographic scan (complements IAS/SCUDO text-layer injection screening).
Scanned documents	OCR errors on low-quality scans propagated as ground truth. Handwriting or degraded text misread by AI. AI analysis built on faulty	Per-region OCR confidence scoring. Low-confidence regions flagged inline — AI claims derived from flagged regions carry a

Modality	Primary failure modes	MAS detection approach
	text extraction.	lower confidence ceiling regardless of model self-reported confidence.

Design Principles

- Runs at media ingestion, before artifacts enter the evidence corpus. Text chunking continues in parallel; MAS screening does not block text processing. High-risk artifacts are flagged in the evidence record before any AI analysis of their content begins.
- Produces a continuous authenticity risk score per artifact (0.0 to 1.0) by modality. Scores are not binary — the goal is to surface signals, not to substitute for human judgment. Thresholds for advisory, flagged, and quarantined dispositions are configurable by PGS/NOMOS policy per media type and matter class.
- Transcripts of video and audio recordings are tagged as derivative artifacts with their own accuracy metadata, separate from the source recording. AI analysis built on a transcript inherits the transcript's OCR or ASR confidence profile, not the authenticity of the original recording. This distinction is surfaced in Source Type Badges: a claim citing a deposition transcript shows TRANSCRIPT rather than GRND.
- MAS does not modify source artifacts. All screening is non-destructive and operates on copies. Source artifacts in the evidence corpus remain unchanged. MAS metadata is attached to the evidence record, not embedded in the artifact.
- Authenticity risk scores propagate into Source Type Badges in ATHENA. A GRND claim sourced from a high-risk-authenticity video shows a combined signal: grounded (source exists and was retrieved) but authenticity-flagged (MAS score above threshold). The two signals are independent and both visible.
- MAS is an architectural sibling to IAS/SCUDO. Both are ingestion-point screening services. IAS screens text chunks for adversarial instruction patterns. MAS screens media artifacts for authenticity signals. They share the ingestion pipeline position and the same scoring, disposition, and quarantine infrastructure pattern.

Multi-Modal Calibration Dimension

CONTEXT

An analyst who is well-calibrated for AI-assisted document analysis may be entirely uncalibrated for AI video analysis. The cognitive task is different, the failure modes are different, and research on automation bias in video review contexts suggests calibration transfers poorly across modalities. MAS provides the per-modality authenticity signal that enables TCS/MIMIR to track analyst calibration separately for video, audio, and image analysis — not just by interaction class, but by the media type underlying the evidence being analyzed.

Service Specification

Namespace	themis-quality alongside IAS/SCUDO, CVS/VERITAS, FGTS/ALETHEIA, TVS/KAIROS, and RQS/HERMES. MAS and IAS are architectural siblings — both are ingestion-point screening services with the same disposition and quarantine infrastructure pattern.

Position in pipeline	At media artifact ingestion, before AI analysis of media content begins. Operates in parallel with text chunking for mixed-media documents. Non-blocking for text processing — MAS risk score is attached to the evidence record asynchronously for standard risk levels; only quarantine disposition is synchronous and blocking.
Detection model registry	Versioned ensemble of classifiers per media type: video deepfake ensemble (facial, temporal, compression), audio synthesis classifier, image manipulation ensemble (GAN, JPEG, metadata), OCR confidence scorer. Model registry is Git-managed with change-controlled updates. AI Governance Committee approval required for classifier threshold changes.
Risk scoring	Continuous 0.0 to 1.0 per artifact per modality. Thresholds configurable by PGS policy: Advisory (0.3-0.6), Flagged (0.6-0.85), Quarantined (>0.85). Practice-group-specific overrides available (e.g., criminal litigation may require lower thresholds than transactional work).
Output per artifact	{artifact_id, modality: video|audio|image|scanned_doc, risk_score, anomaly_signals[], disposition: pass|advisory|flagged|quarantined, screened_at, classifier_versions[]}
Storage	Authenticity event log (PostgreSQL, append-only): MediaAuthenticityEvent per artifact screened. Risk score cache (Redis): keyed on artifact hash, TTL 24 hours. Quarantine registry (Redis): active quarantine state per artifact ID. Transcript accuracy registry (PostgreSQL): per-transcript ASR/OCR confidence metadata.
False positive handling	Legal media artifacts frequently trigger lower-threshold authenticity signals without being manipulated — compression artifacts in old recordings, lighting inconsistencies in surveillance footage, handwriting variation in historical documents. Flagged artifacts are visible to the analyst with specific anomaly signals described. Analysts can release flagged artifacts with an override reason. Override events are logged and reported to AI Governance Committee.

Integration with Existing THEMIS Services

Service	Relationship	Integration detail
Provenance / MOIRAI	Feeds into	MAS writes MediaAuthenticityEvent for every artifact screened. Authenticity risk score and anomaly signals are part of the artifact's provenance record from the moment of ingestion. This feeds the Origin accountability axis: provenance is not only chain of custody but authenticity of the source material.
TVS / KAIROS	Relationship	Authenticity is a validity concern. An artifact later confirmed as manipulated or fabricated should trigger a validity invalidation event propagated through the provenance DAG. MAS produces the initial risk signal. If post-screening investigation confirms manipulation, TVS/KAIROS propagates the invalidation to all evidence records derived from that artifact.
TCS / MIMIR	Feeds into	MAS provides the modality dimension missing from current TCS calibration. Analyst RAI is now tracked separately for video analysis, audio analysis, image analysis, and scanned document analysis. MAS authenticity risk scores are visible to TCS as context for calibration events involving media-derived claims.
IAS / SCUDO	Architectural sibling	Both services screen at ingestion and share the disposition and quarantine infrastructure pattern. IAS screens text for

Service	Relationship	Integration detail
		adversarial instructions; MAS screens media for authenticity signals. They are complementary and non-overlapping in their concern. Image EXIF metadata adversarial injection (text instructions embedded in metadata fields) is screened by MAS at the metadata layer — the one overlap point where both services have partial coverage.
DPS / CODEX	Feeds into	When AI-originated content derived from media evidence is inserted into a document through the authoring solution, the DPS provenance payload includes the MAS authenticity risk score for the source artifact. A work product paragraph grounded in a high-risk-authenticity video carries that signal into the document provenance record and into the finalization snapshot.
ERAS / LOGOS	Feeds into	When the model reasons about media content, ERAS captures both the reasoning chain and the MAS authenticity context of the source. An ERAS reasoning record for a claim derived from video evidence includes the MAS risk score for that video. This is essential for professional responsibility audit: was the AI's reasoning about this media artifact disclosed alongside the fact that the artifact carried an authenticity flag?
FGTS / ALETHEIA	Receives from	When an analyst confirms that a flagged artifact is genuinely manipulated (via Verification Queue dispute action on a media-derived claim), FGTS logs this as a high-quality authenticity ground truth signal. This builds the MAS training corpus over time and informs classifier maintenance.
RQS / HERMES	Relationship	RQS monitors retrieval quality for text chunks. For media artifacts, MAS plays an analogous role: it surfaces quality signals about the artifact before it enters AI analysis. The two services serve the same epistemological function at different data layers.

Epithet Rationale

EIDOLON — in Greek thought, an eidolon was a phantom or apparition: an image that resembled a real person or thing but was not the real thing. The term was used for spirit-doubles, reflections, and deceptive copies. In Homer, the gods send eidola — phantom images — to deceive mortals. For a service whose sole function is to determine whether a media artifact is the real thing or a deceptive copy, EIDOLON is precisely right. The name carries the additional resonance that deepfakes and manipulated evidence are the contemporary form of the ancient problem: something that appears real but is not.

Section 05 — DPS / CODEX

		EPITHET	MEANING
DPS NEW	**Document Provenance Service** *CODEX / Latin for book — Justinian's Codex was the foundational compilation of Roman law.*	**CODEX**	Latin for book — Justinian's Codex was the foundational compilation of Roman law.

Purpose

The THEMIS provenance chain covers session-level and claim-level records through Provenance/MOIRAI. Without a dedicated document layer, that chain severs the moment AI-generated content moves from an ATHENA session into a document production workflow. For firms generating documents at high volume, routing every paragraph-level provenance event directly through Provenance/MOIRAI creates write contention that degrades the core provenance graph for all other services.

DPS is the document-layer service that solves both problems simultaneously. It maintains the chain of custody from ATHENA session through to finalized work product, and it does so through a dedicated buffering and routing layer that absorbs high-frequency document edit events before bulk-flushing to Provenance/MOIRAI. It integrates directly with the firm's custom web authoring solution, which is the production environment for all AI-assisted work product.

Design Principles

- Receives structured provenance payloads from ATHENA session exports. Associates that payload with document content at the paragraph level — not just as document metadata but as inline attribution that persists through the editing lifecycle.
- Manages the complete document lifecycle event stream: AIContentInsertedEvent (when ATHENA-sourced content is placed in a document), ContentEditedEvent (when an attorney modifies AI-originated content, with edit depth scoring), DocumentOpenedEvent (triggering a TVS/KAIROS currency check against all sources underlying AI-originated paragraphs), and DocumentFinalizedEvent (complete provenance snapshot at filing or submission).
- Buffers high-frequency document edit events in Kafka before bulk-flushing to Provenance/MOIRAI. At document generation scale, per-event writes to MOIRAI are not viable. DPS is the buffer layer that makes paragraph-level granularity practical.
- Exposes a real-time query API to the authoring solution's provenance panel. When an attorney is editing a brief, the provenance panel shows the current provenance state of every paragraph — which are AI-originated, which claims are verified, which sources have changed validity since generation. This query path hits DPS's Redis cache, not MOIRAI directly.
- Enforces PCES/AEGIS matter scope at the document composition level. A document drawing on ATHENA sessions from multiple matters requires PCES validation that the combined content does not violate matter boundaries. Cross-matter contamination is as structurally possible in a composed document as in a RAG retrieval.

- Routes edit depth signals to FGTS/ALETHEIA. The semantic distance between AI-generated original text and the human-revised version — computed using the same embedding approach as the ATHENA Reformulation Gate — determines the correction signal weight. A substantively rewritten paragraph carries different epistemic weight than a lightly copyedited one.
- DPS is a THEMIS client of the authoring solution, not the reverse. The authoring solution writes document events to DPS APIs and queries DPS for provenance state. DPS handles all THEMIS service interactions — MOIRAI, TVS, PCES, FGTS — on behalf of the authoring solution. The authoring solution has no direct THEMIS dependencies.

Schema Timing Constraint

CRITICAL

DPS document-level event schemas must be designed in Phase 3-4 alongside Provenance/MOIRAI, not after it. If MOIRAI is built and documents begin generating provenance events before DPS exists, matter records created during that window will lack document-layer provenance that cannot be retroactively reconstructed. The authoring solution integration can deploy in a later phase, but the DPS schema and MOIRAI document event contracts must be established during the core infrastructure build.

Service Specification

Namespace	themis-core alongside Provenance/MOIRAI and TCS/MIMIR. DPS is core infrastructure — the document-layer counterpart to MOIRAI's session and claim layer.
Document registry	PostgreSQL — document records keyed on document ID with paragraph-level provenance maps, version history, edit depth scores, and finalization state.
Write buffer	Kafka — absorbs high-frequency ContentEditedEvents before bulk-flushing to Provenance/MOIRAI. Configurable flush interval and batch size. Replay capability on MOIRAI recovery.
Provenance panel cache	Redis — real-time document provenance state per document ID. Updated on every DPS event. Queried by authoring solution provenance panel on every paragraph focus. TTL: session-length.
Authoring API	Synchronous REST for provenance panel queries (sub-100ms P95 target). Async webhook for document lifecycle event receipt from authoring solution. OpenAPI schema published and versioned.
TVS integration	DocumentOpenedEvent triggers TVS/KAIROS currency check for all source document IDs referenced in the document's provenance map. TVS returns validity state per source. DPS surfaces changed-validity sources as provenance flags in the authoring solution provenance panel.
PCES integration	Document composition requests include the matter scope of all contributing ATHENA sessions. PCES validates the combined scope before AIContentInsertedEvent is accepted. Cross-matter violations are rejected with a specific error code returned to the authoring solution.
Finalization record	DocumentFinalizedEvent generates a complete provenance snapshot: all AI-originated paragraphs with their session source, claim-level provenance, verification states, TVS validity at finalization time, edit depth per paragraph, and TCS/MIMIR RAI snapshot. Published to Provenance/MOIRAI as an immutable document record.

Integration with Existing THEMIS Services

Service	Relationship	Integration detail
Provenance / MOIRAI	Primary downstream	DPS is the document-layer producer for MOIRAI. All document lifecycle events flow from DPS to MOIRAI in batches. MOIRAI does not receive document events directly from the authoring solution. The document-level provenance record at finalization is the canonical MOIRAI entry for the work product.
PCES / AEGIS	Runtime dependency	DPS calls PCES to validate matter scope at document composition time. PCES determines which matter partitions can appear together in a single document. Violations are surfaced to the authoring solution before content is accepted into the document provenance map.
TVS / KAIROS	Runtime query	DPS queries TVS on DocumentOpenedEvent for currency state of all sources referenced in the document. TVS responses are cached in the DPS Redis layer and surfaced in the authoring solution provenance panel. TVS validity change events for active document sources are pushed to DPS via subscription.
FGTS / ALETHEIA	Feeds into	DPS routes ContentEditedEvent with edit depth score to FGTS as a weighted correction signal. Paragraph-level edit depth (embedding distance between AI-generated original and human revision) determines signal weight. Substantially rewritten paragraphs are high-quality correction signals. Light copyedits are low-weight signals.
TCS / MIMIR	Contributes to	DPS includes TCS RAI snapshot at DocumentFinalizedEvent. The analyst's calibration state at the time of document finalization becomes part of the work product provenance record — relevant for professional responsibility audit and matter quality review.
ERAS / LOGOS	Surfaces in authoring	ERAS reasoning captures for AI-originated paragraphs are included in the DPS provenance panel query response. Attorneys editing AI-assisted content can see the reasoning behind claims directly in the authoring solution, without switching to ATHENA.

Epithet Rationale

CODEX — the Latin term for a written legal compilation, most notably Justinian's Codex, the foundational assembly of Roman law that formed the basis for legal systems across the Western world. In THEMIS, DPS is the service that ensures every document produced on the platform carries a complete, ordered record of everything that contributed to it — a codex of the document's intellectual provenance. The name is legible to attorneys, legally resonant, and accurately describes the service function.

Section 06 — TVS / KAIROS Extended Scope

TVS / KAIROS — Scope Extension
Ground truth corpus validity incorporated as a TVS scope extension. No new service required.

Extension Rationale

FGTS/ALETHEIA accumulates a corpus of analyst corrections and ground truth signals that TCS/MIMIR uses to improve calibration over time. That corpus is itself subject to temporal decay. Legal ground truth changes when statutes are amended, courts overturn precedent, or regulatory positions shift. A correction logged in month three of platform operation that was accurate at the time may be misleading by month eighteen — and if TCS continues to treat it as ground truth, calibration degrades rather than improves.

TVS/KAIROS already owns decay functions, active invalidation, and DAG propagation for active matter evidence. The concern for the FGTS corpus is structurally identical: a corpus item has a source, that source has temporal validity, and when the source is superseded the corpus item needs to be flagged. The extension is a scope change and routing addition, not an architectural change. TVS already receives KCS/ARGUS invalidation events — those events now propagate into the FGTS corpus alongside active matter evidence.

What Changes in TVS

- TVS receives a new event subscription: FGTS corpus items register with TVS in the same way active matter chunks do, with their source document references and origination timestamp.
- When a KCS/ARGUS invalidation event arrives for a source that underlies one or more FGTS corpus items, TVS propagates a CorpusItemValidityDecayEvent to FGTS/ALETHEIA alongside the existing evidence validity events.
- FGTS marks affected corpus items as potentially stale and routes them to a human review queue rather than continuing to use them in TCS calibration without qualification.
- Point-in-time reconstruction already available in TVS for active matter evidence is extended to the corpus: TCS can query what the corpus validity state was at any historical moment, enabling audit of calibration quality over time.
- Corpus item decay profiles are configurable separately from evidence decay profiles. Legal ground truth typically has a longer half-life than time-sensitive evidence — decay parameters reflect this difference.

Integration Impact

Service	Relationship	Integration detail
KCS / ARGUS	Unchanged	KCS continues to send ActiveInvalidationRequests to TVS. TVS routing of those events now covers the FGTS corpus in addition to active matter evidence. No KCS changes required.

Service	Relationship	Integration detail
FGTS / ALETHEIA	New consumer	FGTS receives CorpusItemValidityDecayEvents from TVS. Affected corpus items are flagged as potentially stale and routed to human review. FGTS does not remove items from the corpus — it quarantines them from active TCS calibration input pending review.
TCS / MIMIR	Indirect benefit	TCS calibration models are protected from stale ground truth signals. FGTS item staleness flags are visible in the TCS calibration dashboard. Calibration quality metrics now include corpus validity state as a component.
Provenance / MOIRAI	Extended writes	TVS writes CorpusItemValidityDecayEvents to Provenance/MOIRAI alongside existing ValidityChangeEvents. The full corpus validity history is part of the provenance record.

Section 07 — Provenance Chain Cryptographic Attestation

Provenance / MOIRAI — Write-Path Extension + Vault Key Management
No new service. Architectural requirements applied to MOIRAI write path, all event-writing services, and Vault infrastructure.

The Gap

Provenance/MOIRAI is append-only, which prevents modification of events after the fact. Append-only is necessary but not sufficient. Any authorized service can write valid-looking events to an append-only log — a compromised inference gateway, a compromised DPS/CODEX, or a malicious insider with API write access can generate fraudulent provenance records. In a legal context where THEMIS provenance records may be submitted to courts or bar associations as evidence of governed AI use, a chain that any authorized party can write to is not a tamper-evident audit trail. It is a trusted log. The distinction is critical.

Cryptographic event signing closes this gap. Each THEMIS service signs its provenance events with a private key before submission. MOIRAI validates signatures at the write boundary and rejects unsigned or invalidly-signed events. External parties can verify any event using published public keys without accessing THEMIS internal infrastructure.

Three Mechanisms Working Together

Event Signing via Vault

Each THEMIS service has an asymmetric key pair managed by HashiCorp Vault, which is already present in themis-infra. Services never hold private keys directly. To sign an event, a service calls Vault's signing API with the event payload and receives a signature in return. The signature is attached to the event before MOIRAI submission. MOIRAI's write API validates the signature on receipt using the service's published public key. Invalid or missing signatures cause rejection, not logging. Key rotation happens on a defined schedule — quarterly for active-use keys, with longer retention for keys whose events must remain verifiable over multi-year matter timelines. Rotation does not invalidate prior events because the signing key ID is stored with each event and the public key registry retains all historical public keys after rotation.

Hash Chaining

Every event includes a hash of the prior event in the chain. Each new event's hash incorporates its own payload plus the prior event's hash, creating a cryptographic chain where any tampering with a prior event breaks the hash of every subsequent event. An attacker cannot insert, modify, or delete any event in the chain without the break propagating forward and becoming detectable. MOIRAI validates the hash chain on write, rejecting events whose prev_event_hash does not match the actual hash of the prior event.

External Timestamping (RFC 3161)

Every twenty-four hours, MOIRAI submits a digest of the current chain state to an external RFC 3161 compliant timestamp authority and stores the response as an anchor event. This creates a verifiable external reference point independent of THEMIS infrastructure entirely. An external party can verify that the chain existed in a specific state at a specific time without access to THEMIS systems — the TSA token is sufficient. A compromised MOIRAI could forge new events appended after a TSA anchor but could not alter events prior to the anchor without breaking the hash chain. This is the mechanism that makes THEMIS provenance records defensible in adversarial legal contexts.

Event Schema Extension

signature	Cryptographic signature over the canonical event payload. Produced by the originating service via Vault signing API. Stored with the event in MOIRAI.
signing_key_id	Identifies which key pair signed this event. Used to select the correct public key from the registry for verification. Remains valid after key rotation.
prev_event_hash	SHA-256 of the prior event in the service's event stream. Creates the hash chain. MOIRAI validates on write.
tsa_token	RFC 3161 timestamp authority token. Present on TSA anchor events (generated every 24 hours). Absent on regular events — regular events are covered by the nearest subsequent anchor.

Services Required to Sign

Every service that writes provenance events to MOIRAI must implement the Vault signing call. This covers all nine original THEMIS services plus the five new services added in this addendum: CVS/VERITAS, MDS/KRONOS, IAS/SCUDO, MAS/EIDOLON, and DPS/CODEX. The inference gateway signs the context snapshot events it generates for each exchange. The claim extractor signs the claim classification events it produces. No event reaches MOIRAI unsigned.

Verification API Extension

Single event verification	Verify one event's signature, payload integrity, and position in the hash chain. Used for spot audit queries.
Range verification	Verify a sequence of events by walking the hash chain between two event IDs. Confirms no events were inserted, deleted, or modified within the range.
Full chain audit	Verify every event in a session or matter scope, walking hash chains and checking all TSA anchor points. Produces a signed audit certificate suitable for court or bar association submission.
Key registry query	Returns the public key for any signing_key_id, including rotated keys no longer in active use. Allows verification of events signed under any historical key.

PHASE CONST

Cryptographic attestation must be designed and implemented in Phase 3-4 alongside the Provenance/MOIRAI core build. The first analyst interactions with ATHENA are cold start by definition — those initial events will be the foundation of the chain. An unsigned event cannot be retroactively signed without breaking the hash chain. A mix of signed and unsigned events in the same chain provides weaker guarantees than a fully signed chain from inception. There is no viable retrofit path.

RAINT

Section 08 — FGTS Corpus Quality: Correction Confidence Weighting

FGTS / ALETHEIA + TCS / MIMIR — Corpus Quality Extensions

No new service. Correction confidence weighting function in TCS/MIMIR. Human review queue and corpus versioning in FGTS/ALETHEIA.

The Gap

The FGTS ground truth corpus accepts all analyst verification actions as ground truth. Gaming detection (Intervention 25) addresses the behavioral pattern of analysts who satisfy the interface without genuine engagement. It does not address the deeper problem: a sincere, genuinely engaged analyst with a systematically incorrect view of a specific regulatory domain will produce high-engagement, non-gaming corrections that corrupt TCS/MIMIR calibration for that domain across the entire analyst population. Their incorrect verifications enter FGTS indistinguishably from correct ones. High-volume analysts with idiosyncratic views disproportionately shape the corpus. There is no second-opinion mechanism. The current design assumes analysts are calibrated enough to produce useful ground truth — which is precisely what the platform is trying to establish.

Correction Confidence Weighting

Not all corrections are equal. The correction confidence weight for any verification action is a function of five factors. TCS/MIMIR computes this weight at the time each correction event arrives from FGTS and applies it to calibration model updates. Low-weight corrections do not get excluded from the corpus — they get down-weighted, which means they require more volume to shift calibration and are flagged for human review when they contradict higher-weight corrections on the same claim type.

Factor	Source	Effect on weight
Analyst domain RAI	TCS / MIMIR	Domain-specific RAI, not overall RAI. An analyst with a strong overall score but no history in securities litigation is not a high-quality ground truth source for securities litigation claims. Weight scales with domain-specific calibration history.
Session gaming score	TCS / MIMIR (Intervention 25)	Gaming probability score for the session is inverted as a weight multiplier. High gaming score in the session reduces the weight of all corrections from that session. Captures behavioral compliance without epistemic engagement.
Session pressure mode	Session context store (Intervention 26)	Deadline-Critical mode applies a weight multiplier below one. Time-pressured verification is inherently less reliable than deliberate verification. Standard and Elevated modes apply no adjustment.
Supervisory confirmation	DPS / CODEX (Intervention 27)	A supervisory review that confirms or overrides a verification state applies a substantial weight multiplier. Supervisory review is the highest-quality signal in the system. Supervisory override propagates retroactively to adjust the weight of the original correction — a supervisor

Factor	Source	Effect on weight
		noting that a junior's verification was wrong should reduce the original correction's FGTS weight.
Peer agreement	Cross-Analyst Disagreement Detection (Intervention 28)	A second analyst independently making the same verification action on the same claim type in the same domain multiplies both corrections' weights. Two independent analysts reaching the same correction is substantially stronger ground truth than one. Disagreement between analysts sends both corrections to the human review queue rather than accepting either.

Human Review Queue

Corrections whose computed weight falls below a minimum threshold trigger the human review queue in FGTS/ALETHEIA rather than being incorporated into the corpus directly. Three additional trigger conditions route corrections to the queue regardless of computed weight: the correction contradicts a prior high-confidence correction on the same claim type and domain; the claim involved a HIGH confidence badge being marked as Incorrect (the highest-consequence correction type); or the correcting analyst has fewer than the minimum domain-specific verification history required for that claim type.

Queue reviewers are designated knowledge management or quality assurance staff with access to the claim, the source documents, the analyst's correction, their domain calibration history, and any prior corrections on the same claim type. Reviewers accept, reject, or escalate. Accepted corrections enter the corpus at their computed weight. Rejected corrections are logged to Provenance/MOIRAI but not incorporated. Escalated corrections go to a senior practice group attorney. The review queue is surfaced in a Quality Review dashboard accessible to designated reviewers and the AI Governance Committee.

Corpus Versioning and Rollback

FGTS/ALETHEIA maintains versioned snapshots of the corpus at configurable intervals — weekly during active build phases, monthly once stable. If a batch of corrections from an analyst subsequently identified as significantly miscalibrated requires removal, the corpus can be rolled back to the pre-contamination snapshot and recomputed forward excluding those corrections. This is tractable because FGTS uses weighted accumulation rather than gradient descent — removing a defined set of weighted signals and recomputing the accumulation is a defined operation. The rollback capability is tested during Phase 5-6 deployment, not discovered as a requirement during an incident.

Corpus version snapshots are signed events in Provenance/MOIRAI, bringing them within the cryptographic attestation chain defined in Section 07. A corpus snapshot can be verified as having been the actual corpus state at a given point in time, which matters if a firm is ever asked to demonstrate what ground truth data was informing TCS calibration during a specific matter.

What Changes in FGTS and TCS

Service	Relationship	Integration detail
FGTS / ALETHEIA	Internal extension	Human review queue added as a new internal workflow surface. Corpus versioning and snapshot storage added to FGTS storage layer. Correction events extended with weight metadata received from TCS. Supervisory override retroactive weight adjustment wired from DPS/CODEX Intervention 27 supervisory review events.

Service	Relationship	Integration detail
TCS / MIMIR	Internal extension	Correction confidence weighting function added. Receives the five weight inputs per correction event and computes a final weight before calibration model update. Exposes domain-specific RAI scores (separate from overall RAI) as inputs to the weight function. Calibration updates are now weighted-accumulation rather than uniform-accumulation.
Provenance / MOIRAI	New event types	Corpus snapshot events (signed, per Section 07 attestation requirements). Correction weight events (recording the computed weight and contributing factors for each correction). Human review queue decision events (accept, reject, escalate, with reviewer attribution).
DPS / CODEX	Feeds into	Supervisory review confirmations and overrides from Intervention 27 trigger retroactive weight adjustment events in TCS/MIMIR for the original correction. DPS is the integration point because it manages multi-analyst session attribution.
Intervention 28	Feeds into	Cross-analyst peer agreement signals are routed from the disagreement detection mechanism into the FGTS weighting function as peer agreement multipliers. Cross-analyst disagreement on the same claim sends both corrections to the human review queue.

Section 09 — Updated Integration Map

The following updates the Service Dependency Matrix from Section 05 of the original THEMIS Platform Architecture to include the five new services, the TVS scope extension, the MOIRAI cryptographic attestation extension, and the FGTS/TCS corpus quality extensions.

Service	Depends on	Feeds into
CVS / VERITAS	PGS/NOMOS (interaction class), PCES/AEGIS (matter scope), Provenance/MOIRAI (turn record)	ERAS/LOGOS (unsupported claim signal), FGTS/ALETHEIA (fabrication detection), Provenance/MOIRAI (citation verification events — signed)
MDS / KRONOS	Vendor API endpoints (version polling), Provenance/MOIRAI (active session records)	Provenance/MOIRAI (version change events — signed), TCS/MIMIR (calibration re-validation trigger), AI Governance Committee (notification)
IAS / SCUDO	RQS/HERMES (pre-ingestion position), PGS/NOMOS (quarantine policy)	RQS/HERMES (sanitised chunks), PGS/NOMOS (injection event log), Provenance/MOIRAI (adversarial input events — signed)
MAS / EIDOLON	Provenance/MOIRAI (artifact records), PGS/NOMOS (disposition thresholds), TVS/KAIROS (invalidation trigger on confirmed manipulation)	Provenance/MOIRAI (MediaAuthenticityEvents — signed), TCS/MIMIR (modality calibration dimension), FGTS/ALETHEIA (authenticity ground truth signals), DPS/CODEX (artifact risk scores in document provenance), ERAS/LOGOS (authenticity context in reasoning captures)
DPS / CODEX	PCES/AEGIS (matter scope), TVS/KAIROS (currency queries), Provenance/MOIRAI (document event target)	Provenance/MOIRAI (document lifecycle events — signed and batched), FGTS/ALETHEIA (edit depth correction signals), TCS/MIMIR (RAI snapshot at finalization)
TVS/KAIROS (extended)	Provenance/MOIRAI, KCS/ARGUS (invalidation events) — unchanged	FGTS/ALETHEIA ground truth corpus (new), RQS/HERMES, KCS/ARGUS — existing feeds unchanged
Provenance/MOIRAI (crypto ext.)	Vault (signing key validation), all event-writing services (signature on every incoming event), RFC 3161 TSA (external timestamp anchors)	All consumers of MOIRAI events (now receive signed, hash-chained events). Verification API extended for external audit use.
FGTS/ALETHEIA (quality ext.)	TCS/MIMIR (correction weight computation), DPS/CODEX (supervisory override signals), Intervention 28 (peer agreement signals)	TCS/MIMIR (weighted corrections), Provenance/MOIRAI (corpus snapshot events — signed, review queue decisions), AI Governance Committee quality review dashboard

NOTE The original nine-service dependency matrix remains unchanged for all existing service relationships. This table shows only the new and modified rows. The complete updated matrix is available as a separate appendix on request.

Updated Namespace Assignments

Namespace	Services	Note
themis-gates	PCES/AEGIS, PGS/NOMOS	Unchanged from original architecture.
themis-core	Provenance/MOIRAI, TCS/MIMIR, DPS/CODEX (new)	DPS added to this namespace. It is core infrastructure — the document-layer counterpart to Provenance/MOIRAI. Both must be built concurrently in Phase 3-4.
themis-quality	FGTS/ALETHEIA, TVS/KAIROS, RQS/HERMES, CVS/VERITAS (new), IAS/SCUDO (new), MAS/EIDOLON (new)	CVS, IAS, and MAS added to this namespace. All three are quality and authenticity assurance services operating on model output, retrieval input, and media artifact ingestion respectively.
themis-intelligence	KCS/ARGUS, ERAS/LOGOS, MDS/KRONOS (new)	MDS added to this namespace. All three services are external monitoring or reasoning intelligence functions operating at the platform intelligence layer.
themis-infra	Kafka, Kong, Vault	Unchanged. MDS and CVS add new external API credential entries to Vault.

Section 10 — Updated Implementation Roadmap

The following updates Sections 09 and the investment summary of the original THEMIS Platform Architecture. Phase 1-2 is unchanged. CVS, MDS, DPS, and cryptographic attestation are added to Phase 3-4. IAS, MAS, and FGTS quality extensions are added to Phase 5-6. TVS scope extension is incorporated into Phase 7-8.

1-2	Wks 1-8	**PCES, PGS**	• Safety gates unchanged. PCES and PGS deployed as specified in original architecture. • No new services in this phase.

3-4	Wks 9-28	**Provenance, TCS, CVS, MDS, DPS + Crypto**	• Provenance/MOIRAI and TCS/MIMIR deployed as specified in original architecture. • Cryptographic attestation: Vault key pairs provisioned per service, event schema extended with signature/prev_event_hash/tsa_token fields, MOIRAI write-path signature validation live, RFC 3161 TSA integration operational, verification API extended for range and full-chain audit, all services updated to sign events via Vault before MOIRAI submission. • CVS/VERITAS: citation pattern library built, Westlaw and Lexis API integrations established, Not Found interrupt UI deployed in ATHENA, CitationVerificationEvents signed and published to Provenance. • MDS/KRONOS: vendor API polling configured, ModelVersionChangeEvents signed and published, matter pinning UI deployed in AI Governance Committee dashboard. • DPS/CODEX: document-level event schema designed in concert with MOIRAI schema, Kafka write buffer and Redis provenance cache deployed, authoring solution API contract published, all DPS events signed via Vault. • End of Phase 4: fully signed and hash-chained provenance graph from day one. Complete chain of custody from ATHENA session through finalized work product. Generation-time citation verification and model version change visibility live.

5-6	Wks 29-46	**FGTS, TVS, RQS, IAS, MAS + FGTS Quality**	• FGTS/ALETHEIA and TVS/KAIROS and RQS/HERMES deployed as specified in original architecture. • FGTS quality extensions: correction confidence weighting function in TCS/MIMIR live, human review queue deployed in FGTS and surfaced in Quality Review dashboard, corpus versioning and snapshot storage operational, supervisory override retroactive weight adjustment wired from DPS/CODEX, peer agreement multiplier wired from Cross-Analyst Disagreement Detection (Intervention 28), corpus snapshot events signed per cryptographic attestation requirements. • IAS/SCUDO: injection taxonomy v1.0 built and reviewed by AI Governance Committee, ingestion pipeline integration deployed, chunk risk scoring live, quarantine

			registry operational. • MAS/EIDOLON: detection model ensemble v1.0 built per modality, media ingestion pipeline integration live, authenticity risk scores surfacing in ATHENA Source Type Badges. • End of Phase 6: quality feedback loops running with weighted correction confidence. Retrieval quality monitored. Adversarial ingestion and media authenticity screening active. FGTS corpus quality governed.
7-8	Wks 47-66	**KCS, ERAS, TVS extension**	• KCS/ARGUS and ERAS/LOGOS deployed as specified in original architecture. • TVS/KAIROS scope extended: FGTS corpus item registration with TVS, CorpusItemValidityDecayEvent schema published, FGTS human review queue for stale corpus items operational, point-in-time corpus validity reconstruction available. • End of Phase 8: all thirteen services operational across all three accountability axes. Full platform metrics dashboard live.

Revised Service Count

Original architecture	This addendum	Total
9 services	**+ 5 new services**	**= 13 services**
PCES, PGS, Provenance, TCS, FGTS, TVS, RQS, KCS, ERAS	CVS/VERITAS, MDS/KRONOS, IAS/SCUDO, MAS/EIDOLON, DPS/CODEX (TVS, MOIRAI, FGTS extended — not new services)	TVS scope: ground truth corpus validity. MOIRAI extension: cryptographic attestation. FGTS extension: corpus quality weighting. Document and media provenance chain complete.

End of Addendum A | Five new services | Three service extensions | THEMIS total: 13 services